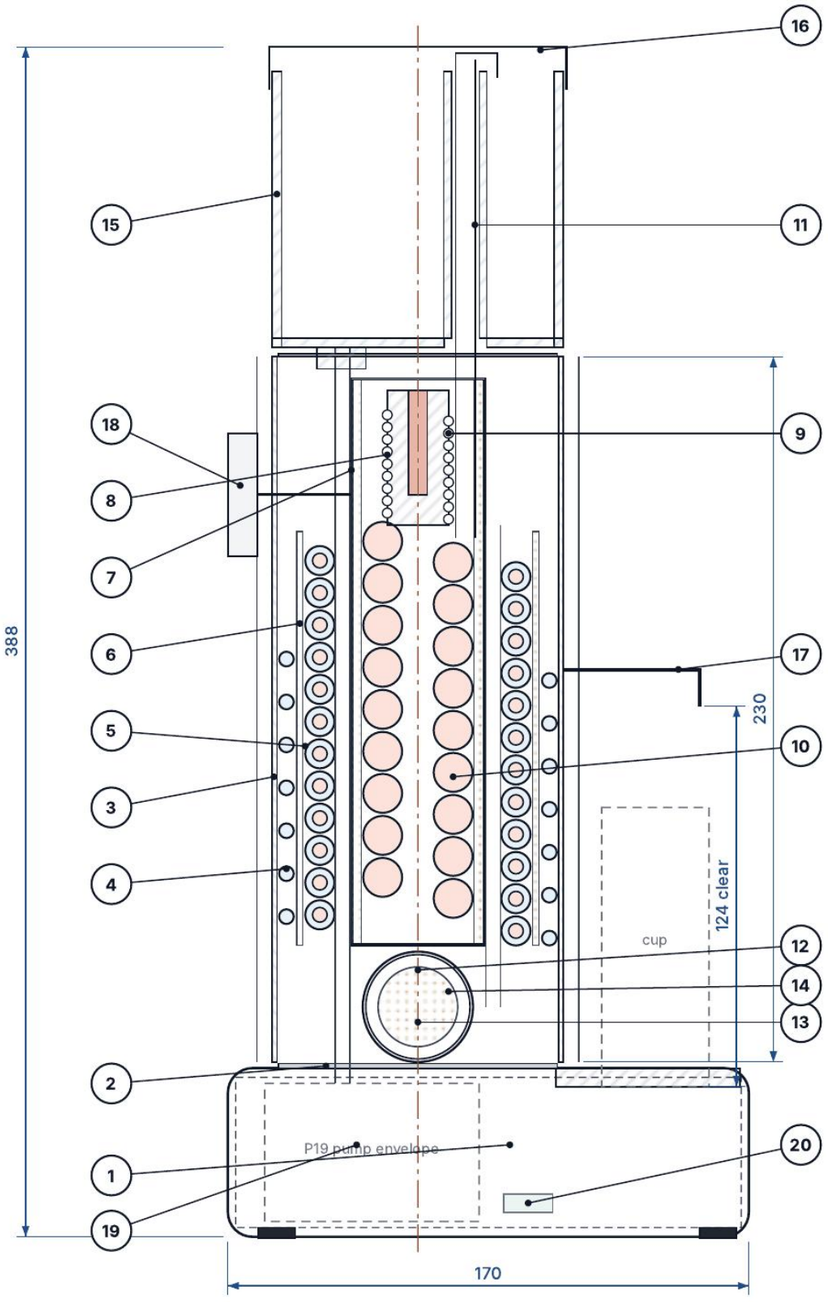


SECTION A-A (1:2)

PARTS LIST

#	Part no.	Name	Material / spec	Thk.	Qty	M/B	g
1	MPx-BLM-001	Base housing	PC/ABS UL94 V-0 (proto ASA)	wall 2.5	1	Make	250
2	MPx-BLM-002	Heat-shield plate	Al 5052-H32 + 0.5 mica	t 1.5	1	Make	26
3	MPx-BLM-003	Shell, finned	Al 6063-T5 extrusion, black anod.	wall 1.5	1	Make	363
4	MPx-BLM-004	Body (cooling) coil	316L tube 3/16" × 0.020"	0.51 wall	1	Make	109
5	MPx-BLM-005	Exchanger coil	316L 3/16" in 3/8" tube-in-tube	0.51 / 0.71	1	Make	608
6	MPx-BLM-006	Insulation sleeve	Silicone foam, closed-cell	t 2.0	1	Make	8
7	MPx-BLM-007	Core can + lid	304 deep-drawn + 3 mm aerogel	t 0.6	1	Make	137
8	MPx-BLM-008	Heater block	Al 6061-T6 bar Ø25	—	1	Make	50
9	MPx-BLM-009	Mini-tube heater coil	316L tube 1/8" × 0.020"	0.51 wall	1	Make	21
10	MPx-BLM-010	Boil coil	316L tube 1/2" × 0.035"	0.89 wall	1	Make	172
11	MPx-BLM-011	Vent standpipe	316L tube 1/4" × 0.028"	0.71 wall	1	Make	30
12	MPx-BLM-012	Lime-trap drawer body	316L tube Ø36 × 1 + ports	t 1.0	1	Make	70
13	MPx-BLM-013	Catcher cartridge	316L 25 µm Dutch weave mesh	—	1	Make	25
14	MPx-BLM-014	Wax gate valve	Wax thermal actuator 92–95 °C	Ø22 × 30	1	Buy	25
15	MPx-BLM-015	Jug with chimney	Borosilicate 3.3 glass	wall 3.0	1	Buy*	294
16	MPx-BLM-016	Cap	304 spun (proto: printed ASA)	t 0.8	1	Make	50
17	MPx-BLM-017	Spout	316L 3/16" tube + nozzle	—	1	Make	15
18	MPx-BLM-018	Dial thermometer	Bimetal Ø40, 0–120 °C, 63 stem	—	1	Buy	40
19	MPx-BLM-019	Pump	Kamoer KPP2 24 V, PharMed tube	—	1	Buy	180
20	MPx-BLM-020	Electrics kit	PD trigger, 20 V 60 W heater, KSD301, TCO	see sh. 8	1	Buy	60
21	MPx-BLM-021	Top plate	304 sheet	t 1.0	1	Make	52

Total dry mass ≈ 2.73 kg (incl. fittings, seals, insulation, spout, dial ≈ 260 g). * glassblower to order.

Note: this supersedes the ~1.3 kg on the v0.5 concept sheet; the heat-exchanger tubes and shell dominate.

FLOW PATH

Jug 15 → pump 19 → exchanger 5, outer tube, upward (25 → ~95 °C)

→ mini-tube heater 9 on block 8 (→ ~100 °C) → boil coil 10 (~60 mL, ~3 min boiling)

vent standpipe 11: steam + CO_2 out; overflow back to jug during warm-up

→ drawer 12: catcher 13 (25 µm) → wax gate 14 (opens ≥ 92–95 °C) → riser

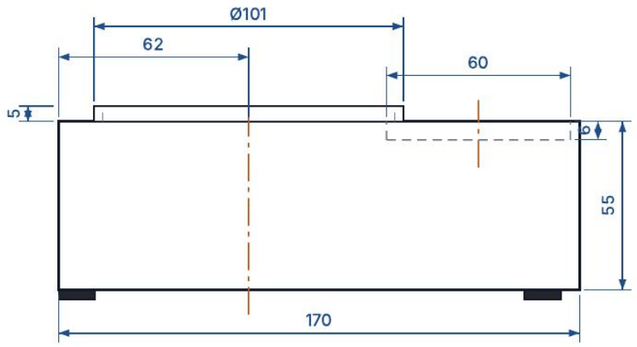
→ exchanger 5, inner tube, downward (→ ~29 °C) → body coil 4 on shell 3 (→ ~27 °C) → spout 17

NOTES

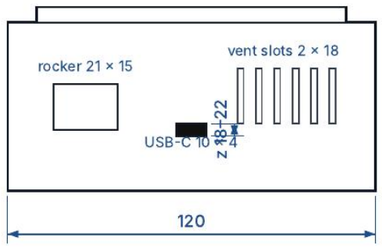
1. Section A-A through body axis; spout side to the right.
2. Water path wetted parts: 316/316L stainless or borosilicate 3.3 only.
3. Prototype joints: twin-ferrule compression; production: brazed (Ag-free, food-safe).
4. Torque compression nuts 1¼ turns from finger-tight; leak test 20 kPa, 10 min.

MPx-BLM-000 · General assembly

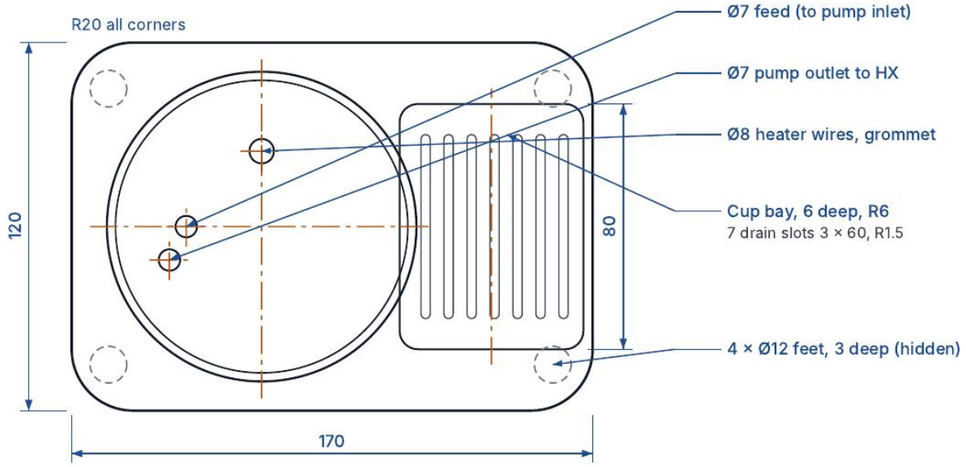
MATERIAL see parts list	FINISH see parts list	SCALE 1:2
QTY / UNIT 1	MASS (est.) ~2.7 kg dry	UNITS · TOL. mm · ISO 2768-m
PROJECTION First angle (IS/ISO)	DRAWN MPx team (PD-298)	DATE 21 Oct 2026



FRONT VIEW (1:2)

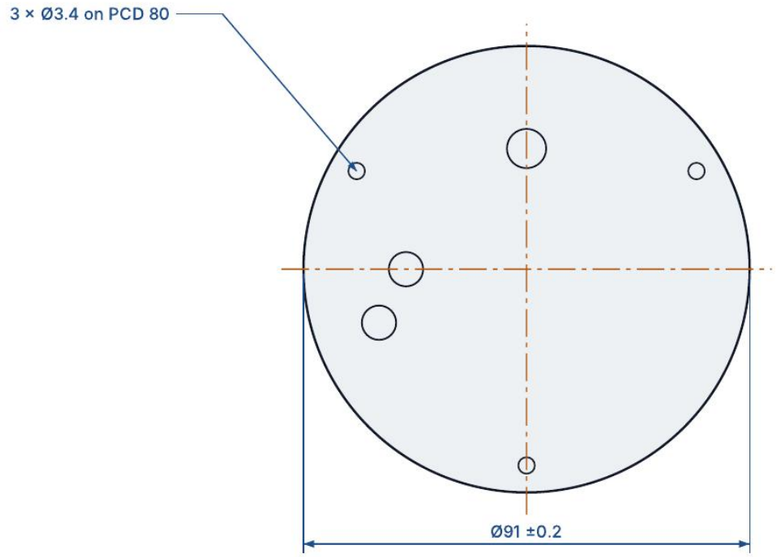


REAR VIEW (1:2)



PLAN VIEW (1:2)

P02 HEAT-SHIELD PLATE (0.8 : 1)



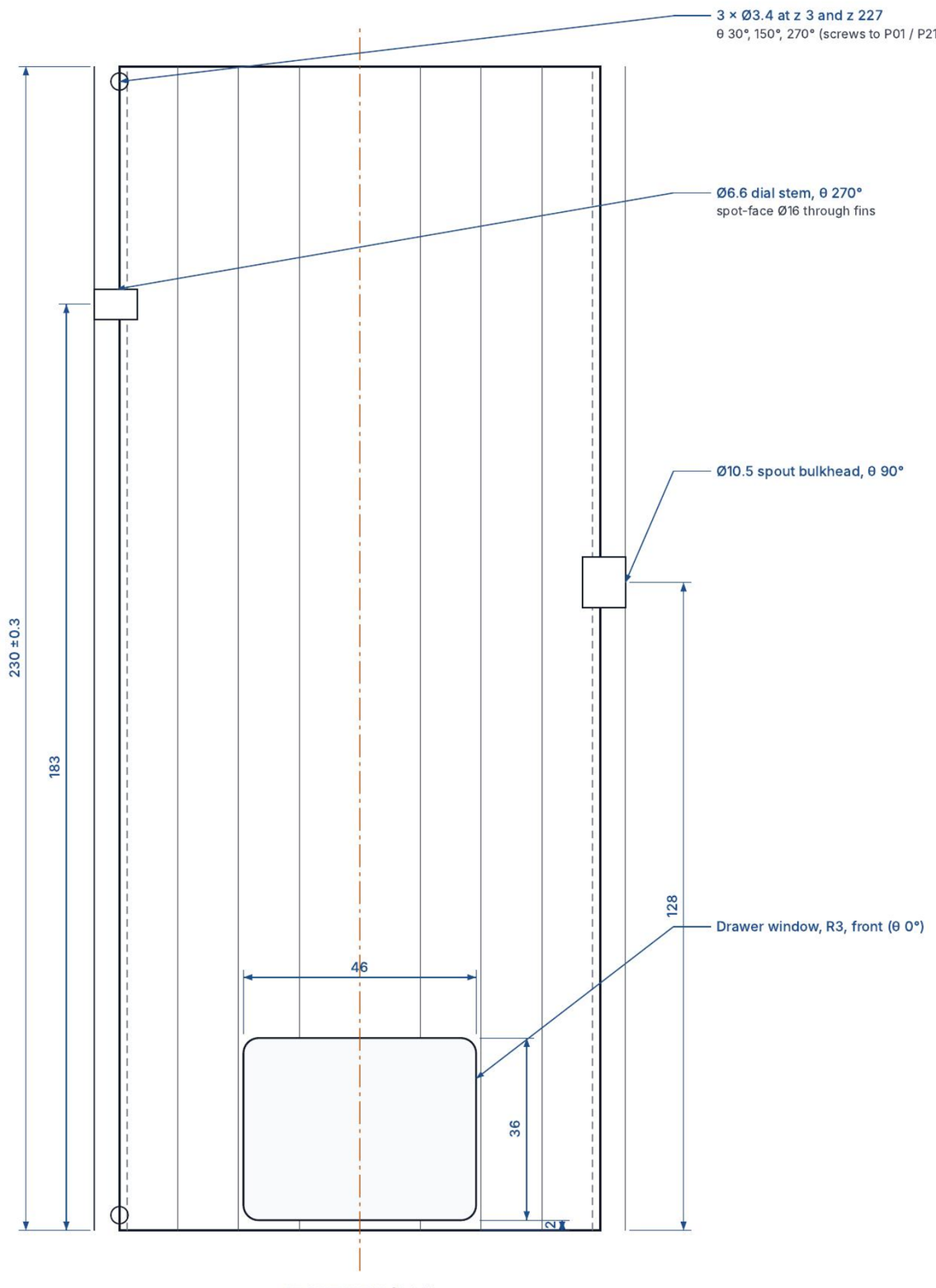
Al 5052-H32 t 1.5 + mica 0.5; holes match P01 deck

NOTES

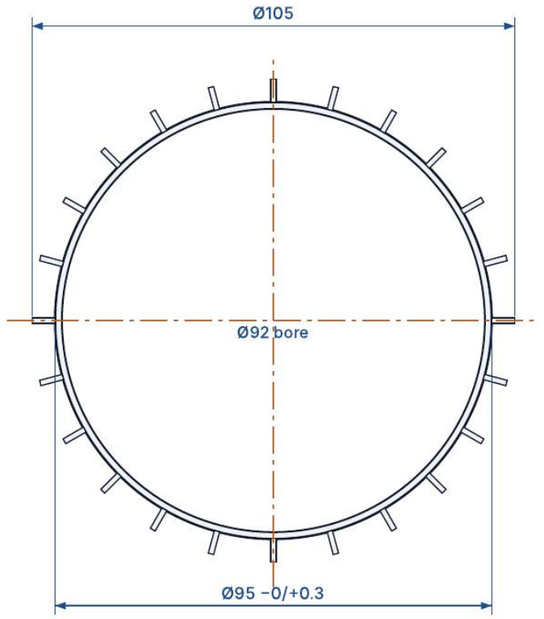
1. Prototype P01: FDM ASA, 0.2 layer, 4 perimeters, 30% gyroid; production: moulded PC/ABS, draft 1°.
2. Wall 2.5, top deck 3.0, internal ribs 1.8 at 30 pitch under the deck. Fillets R1 unless noted.
3. Bottom cover (not shown): 2.0 PC/ABS, 6 × M3 self-tapping, IP54 gasket 1.5 EPDM cord.
4. P02 sits on the deck inside the ring; mica 0.5 bonded underneath (high-temp silicone).

MPx-BLM-001 / 002 · Base housing + heat-shield plate

MATERIAL	FINISH	SCALE
PC/ABS UL94 V-0 · Al 5052-H32	Textured black · plate: plain	1:2 · plate 0.8:1
QTY / UNIT	**MASS (est.)**	**UNITS · TOL.**
1	250 g · 26 g	mm · ISO 2768-m
PROJECTION	**DRAWN**	**DATE**
First angle (IS/ISO)	MPx team (PD-298)	21 Oct 2026

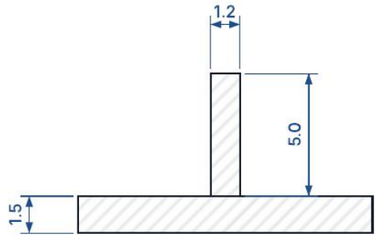


FRONT VIEW (θ 0°)



END VIEW (0.75 : 1)

24 fins at 15°, root to tip 5.0



DETAIL A (4 : 1)

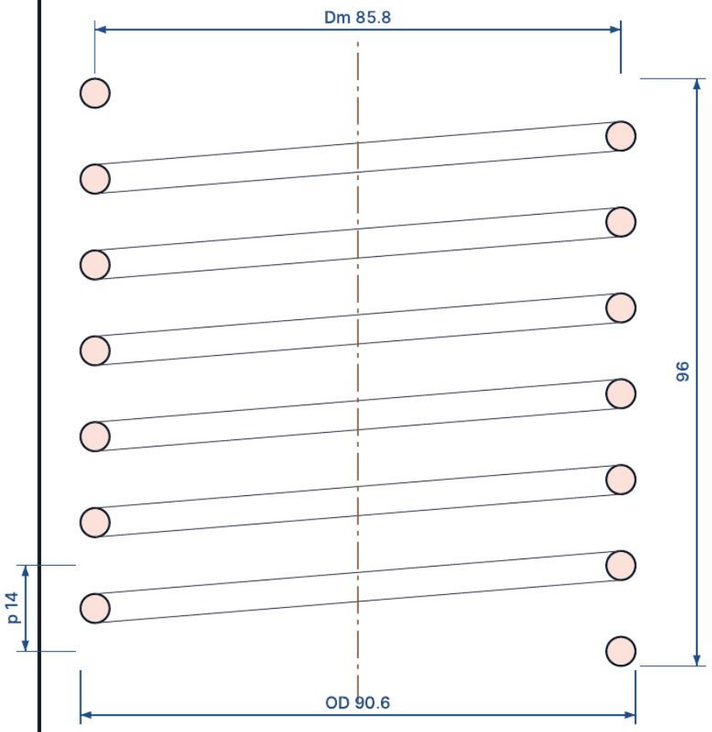
fin root R0.3, tip R0.3

NOTES

1. Extrusion profile Ø95 × 1.5 wall with 24 radial fins 5 × 1.2; cut to 230 ±0.3, deburr, ends square to 0.1.
2. Fins machined off locally (spot-face Ø16, depth to tube OD) at the spout and thermometer holes.
3. Body coil P04 bonded to the bore with thermally conductive epoxy (≥ 1.5 W/m·K), 0.1–0.2 bond line.
4. Prototype: plain 6063 tube Ø95 × 1.5 with clamp-on fin strip is acceptable.

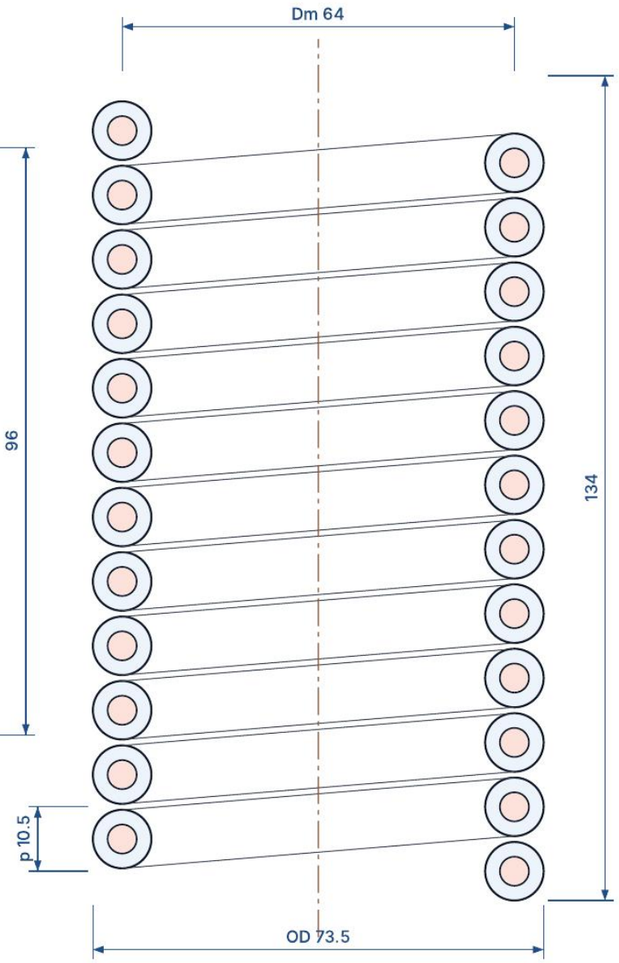
MPx-BLM-003 · Shell, finned

MATERIAL Al 6063-T5 extrusion (EN 755)	FINISH Black anodise Type II, 15–20 µm	SCALE 1:1 · detail 4:1
QTY / UNIT 1	MASS (est.) 363 g	UNITS · TOL. mm · ISO 2768-m
PROJECTION First angle (IS/ISO)	DRAWN MPx team (PD-298)	DATE 21 Oct 2026



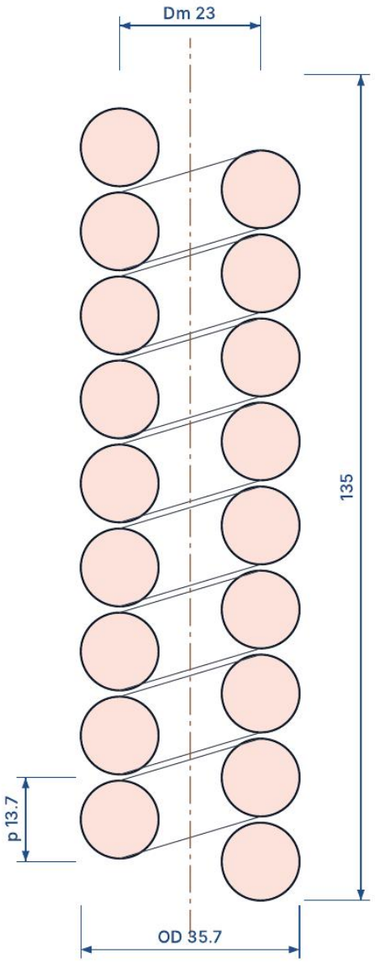
P04 BODY (COOLING) COIL

3/16" × 0.020" (4.76 × 0.51) 316L
6.5 turns · helix 1754 + leads = 2000 cut
lower lead → HX hot outlet tee
upper lead → spout bulkhead (z 185)



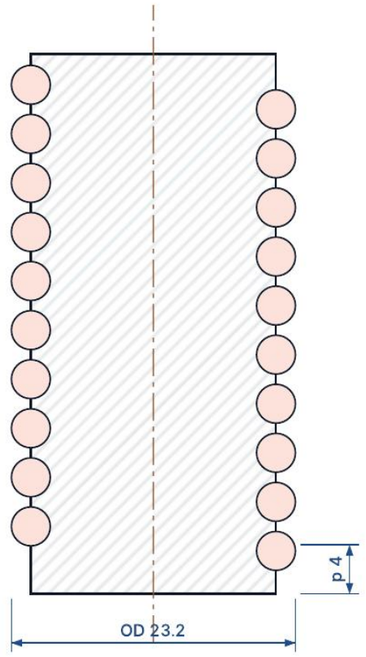
P05 EXCHANGER COIL (TUBE-IN-TUBE)

outer 3/8" × 0.028" (9.53 × 0.71) 316L, 2400 cut
inner 3/16" × 0.020" (4.76 × 0.51), 2750 cut
11.9 turns · helix 2396
2 bored-through tees 3/8" run × 1/4" branch
cold: in bottom tee → out top tee (annulus)
hot: in top → out bottom (inner tube)



P10 BOIL COIL

1/2" × 0.035" (12.7 × 0.89) 316L
8.9 turns · helix 655, 650 cut
internal volume ≈ 61 mL (≈ 3 min at 1.2 L/h)
top: tee to standpipe P11; bottom → drawer



P09 MINI-TUBE HEATER COIL (2:1)

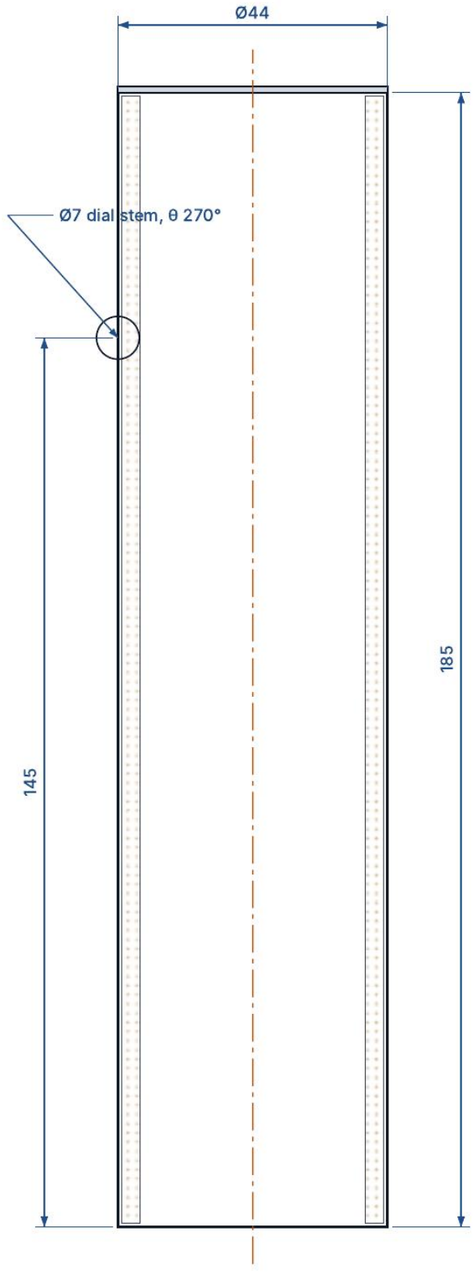
1/8" × 0.020" (3.18 × 0.51) 316L, 600 cut
9.5 turns in the R1.65 groove of P08
pressed in + thermal paste

NOTES

1. Wind on a mandrel of (Dm − OD) diameter; tube filled with dry sand or a bending spring to prevent ovality > 5%.
2. Tube-in-tube (P05): insert 3/16" into 3/8" straight, then wind together; keep the inner tube centred with 3 wire spacers per turn.
3. Leads: 40 straight at each end unless stated; cut square, deburr, flush with citric acid 5% after forming.
4. Helix hand: right-hand for all coils. Pitch tolerance ±1 per turn, Dm ±1.

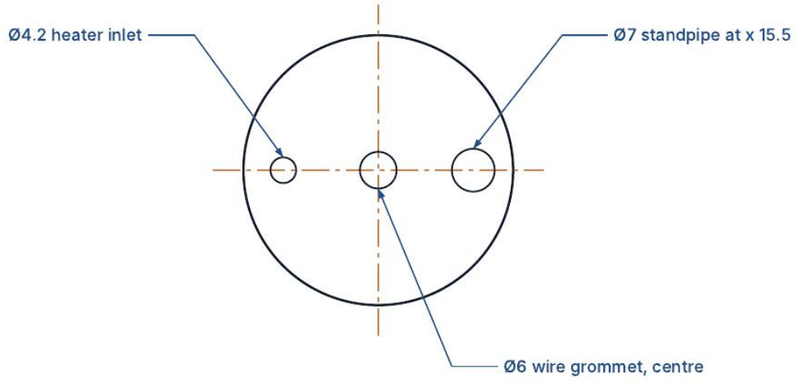
MPx-BLM-004/005/009/010 · Coils (cooling, exchanger, heater, boil)

MATERIAL	FINISH	SCALE
316L seamless tube, ASTM A269	Pickled + passivated, ASTM A967	1:1 · P09 2:1
QTY / UNIT	MASS (est.)	UNITS · TOL.
1	909 g	mm · ISO 2768-m
PROJECTION	DRAWN	DATE
First angle (IS/ISO)	MPx team (PD-298)	21 Oct 2026

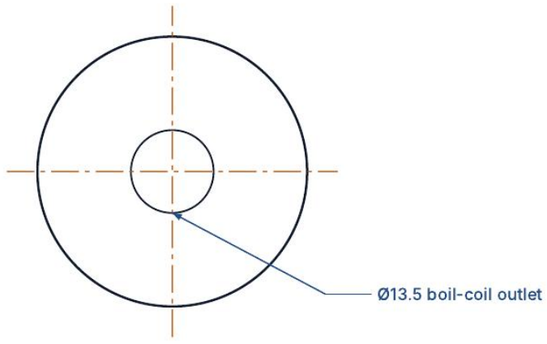


P07 CORE CAN (1:1)

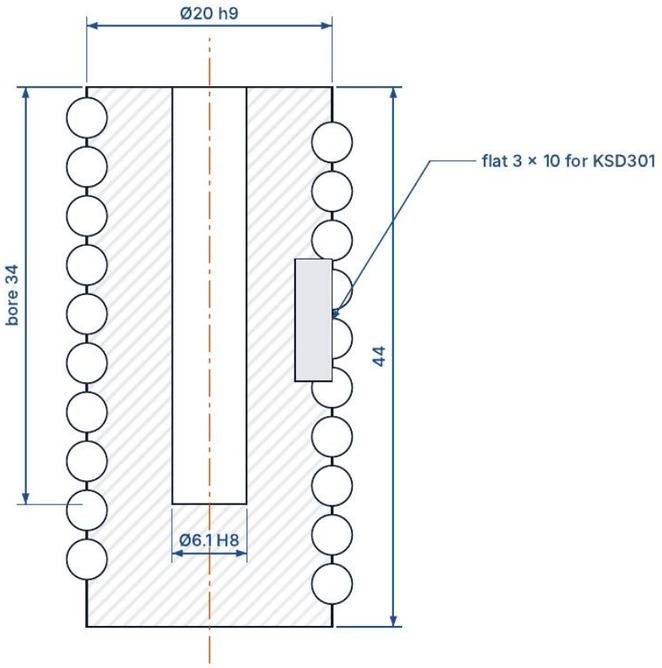
304 t 0.6 · aerogel liner 3.0



LID (from above)

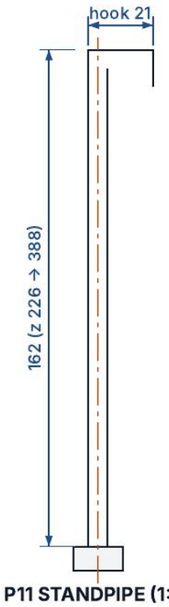


BOTTOM (from below)



P08 HEATER BLOCK — SECTION (2:1)

Al 6061-T6 bar Ø22, turned; groove R1.65 RH pitch 4



P11 STANDPIPE (1:2)

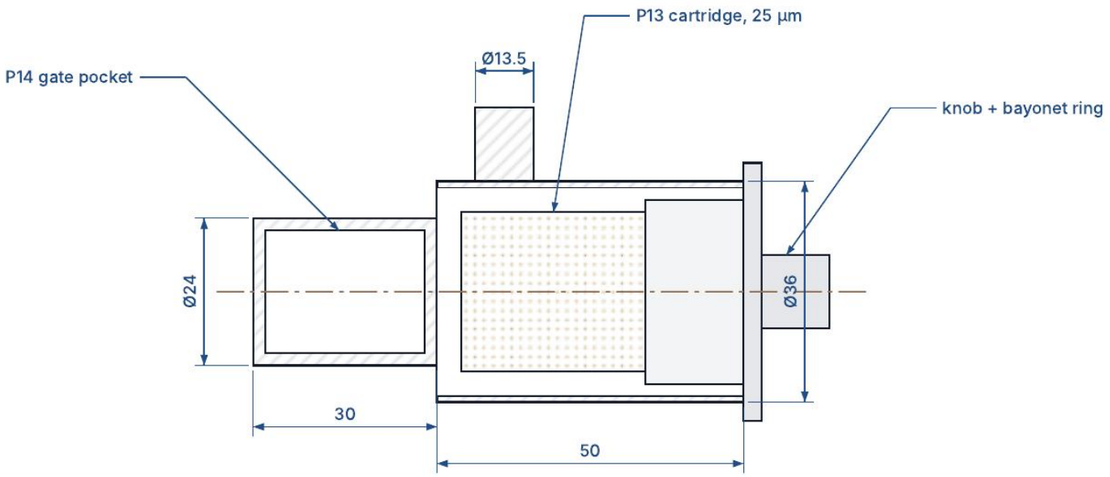
1/4" × 0.028" 316L

NOTES

1. P07 can deep-drawn 0.6 or rolled + seam-welded; lid 0.6 press-fit, 3 tabs. Line inside with 3.0 aerogel blanket.
2. P08 bore Ø6.1 H8 for a Ø6 × 30 cartridge; groove R1.65 helical, pitch 4, 9.5 turns, depth 1.6.
3. P11 standpipe vents steam/CO_2 and returns warm-up overflow to the jug via the hook at the top.
4. KSD301 130 °C clipped to P08 flat (3 × 10 milled), TCO 152 °C in series, bonded with Kapton + paste.

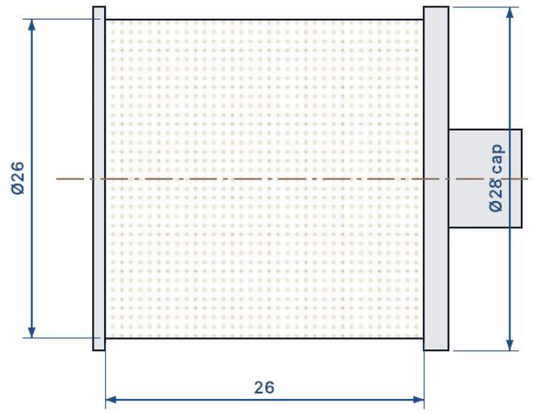
MPx-BLM-007/008/011 · Core can, heater block, vent standpipe

MATERIAL	FINISH	SCALE
304 sheet · Al 6061-T6 · 316L tube	Can 2B; block machined; passivated	1:1 · block 2:1
QTY / UNIT	**MASS (est.)**	**UNITS · TOL.**
1	137 g · 37 g · 30 g	mm · ISO 2768-m
PROJECTION	**DRAWN**	**DATE**
First angle (IS/ISO)	MPx team (PD-298)	21 Oct 2026



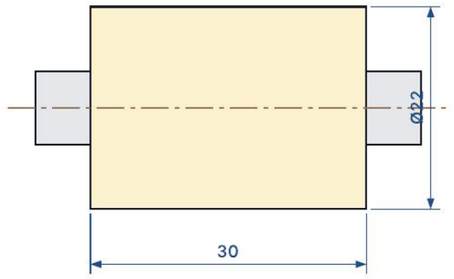
P12 DRAWER — SECTION ON AXIS (1:1)

front (+y) to the right; flow: top port → mesh → core → gate



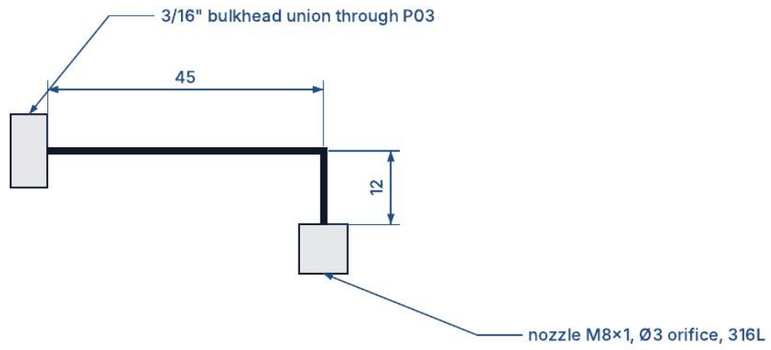
P13 CATCHER CARTRIDGE (2:1)

mesh area ≈ 21 cm² · rinse after each 3–5 L



P14 WAX GATE (envelope)

opens 92–95 °C · lift 5 · Ø8 poppet



P17 SPOUT (1:1)

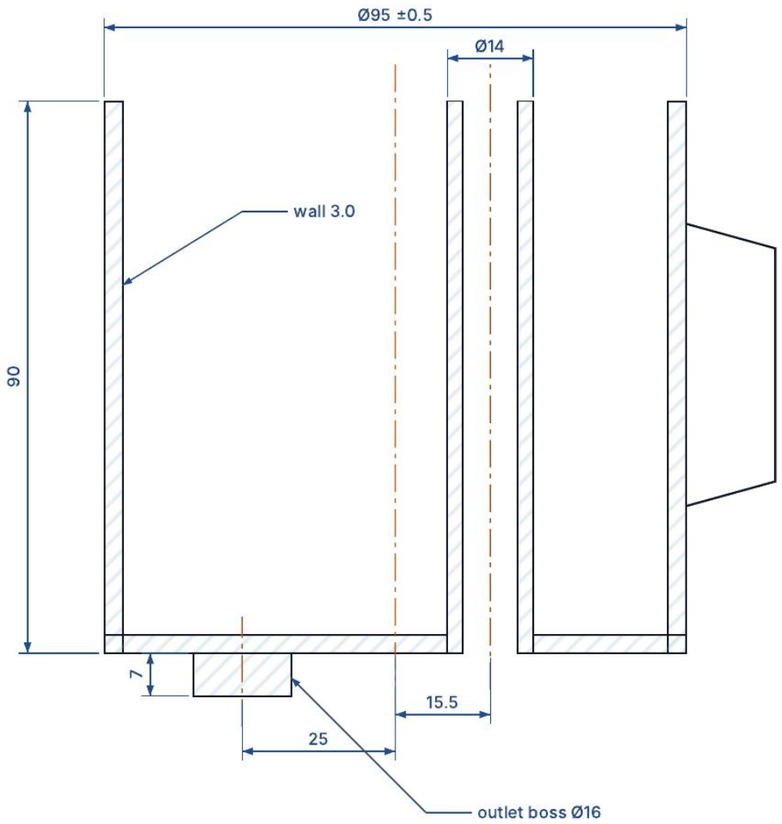
3/16" × 0.020" 316L, bend R12, detachable for cleaning

NOTES

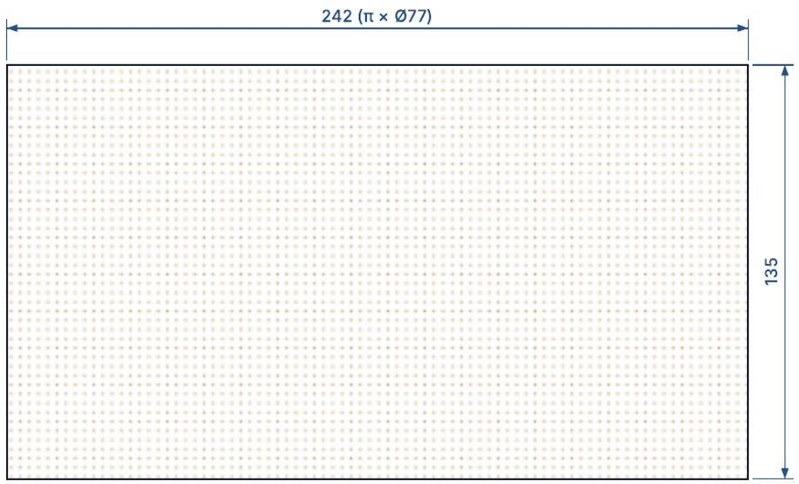
1. P12 body: 316L tube Ø36 × 1.0 × 50; ports TIG-welded: top Ø13.5 boss (boil-coil), rear Ø24 gate pocket.
2. Front bayonet ring 316L, 2 lugs at 180°, ¼ turn; face seal O-ring 30 × 2.5 silicone (VMQ, FDA).
3. P13: 25 µm twilled Dutch weave (165 × 1400) rolled, seam spot-welded, end caps 0.5 316L; knob 316L.
4. P14: wax thermal actuator, opens 92–95 °C, full lift 5 at 100 °C, poppet Ø8 with VMQ seat; buy-in.

MPx-BLM-012/013/014/017 · Lime-trap drawer, catcher, gate, spout

MATERIAL 316L tube + bar · 316L mesh · wax actuator	FINISH Passivated; TIG back-purged	SCALE 1:1 · cartridge 2:1
QTY / UNIT 1	MASS (est.) 70 · 25 · 25 · 15 g	UNITS · TOL. mm · ISO 2768-m
PROJECTION First angle (IS/ISO)	DRAWN MPx team (PD-298)	DATE 21 Oct 2026

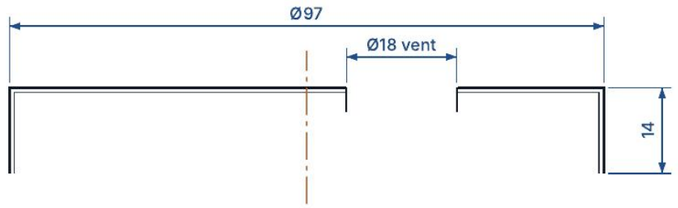


P15 JUG — SECTION (1:1)



P06 INSULATION SLEEVE — FLAT PATTERN (1:2)

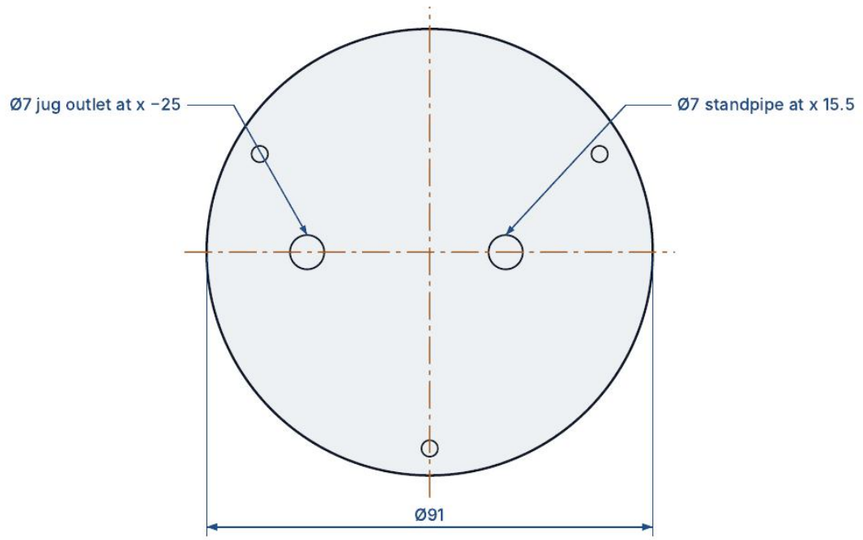
closed-cell silicone foam t 2.0; wraps P05 (OD 73.5) to Ø77.5



P16 CAP (1:1)

304 spun t 0.8 (proto: ASA print)

P21 TOP PLATE (0.8 : 1)



304 t 1.0; 3 × Ø3.4 PCD 80 to P03

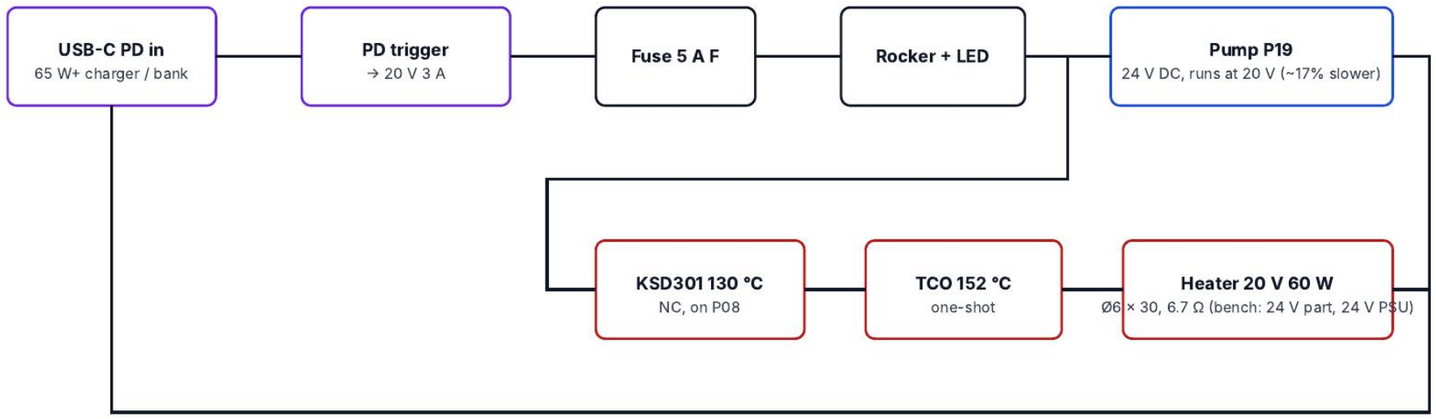
NOTES

1. P15 by scientific glassblower: wall 3.0 ±0.3, chimney fused at x 15.5, outlet boss Ø16 at x −25 for a 1/4" SS bulkhead + VMQ washers.
2. Usable volume to the 0.5 L mark ≈ 0.53 L; graduations every 100 mL (enamel print).
3. P21 top plate carries the jug on a 2.0 VMQ gasket ring; P16 cap is a loose lid with a Ø18 vent over the chimney.
4. P06 sleeve: closed-cell silicone foam 2.0, wrapped round P05 and butt-joined with silicone tape.

MPx-BLM-015/016/021/006 · Jug, cap, top plate, insulation sleeve

MATERIAL	FINISH	SCALE
Borosilicate 3.3 · 304 · 304 · silicone foam	Glass annealed; steel 2B / brushed	1:1 · sleeve 1:2
QTY / UNIT	MASS (est.)	UNITS · TOL.
1	294 · 50 · 52 · 8 g	mm · ISO 2768-m
PROJECTION	DRAWN	DATE
First angle (IS/ISO)	MPx team (PD-298)	21 Oct 2026

WIRING SCHEMATIC



Production cartridge wound for 20 V 60 W (6.7 Ω): ~22 W steady load, warm-up ≈ 15 min. A stock 24 V 60 W part gives only ~42 W at 20 V, so bench-test it on a 24 V PSU.

Seal / item	Where	Material	Size
O-ring	drawer face seal	VMQ (FDA 21 CFR 177.2600)	30 × 2.5
Gasket ring	jug on top plate	VMQ sheet	Ø91/Ø80 × 2.0
Washers ×2	jug outlet bulkhead	VMQ	Ø16/Ø6.5 × 1.5
Grommet	core lid wires	VMQ	Ø6 hole
Twin-ferrule fittings	all tube joints (proto)	316 body + 316 ferrules	1/8", 3/16", 1/4", 3/8", 1/2"
Screws	shell–base, shell–top plate	A4 (316) stainless	6 × M3 × 6 + rivnuts
Screws	base cover	A2 self-tapping	6 × M3 × 10

Material	Grade / standard	Used for	Why
Stainless steel	316L, ASTM A269 tube; 316 bar	all hot wetted tube, drawer, mesh	inert, citric-acid descalable
Stainless steel	304 sheet (EN 1.4301)	core can, top plate, cap	dry or splash only
Aluminium	6063-T5 (shell), 6061-T6 (block), 5052	shell, heater block, shield	conducts heat, light
Glass	Borosilicate 3.3 (ISO 3585)	jug	inert, thermal shock 160 K
Polymer	PC/ABS UL94 V-0 (proto ASA)	base, cover	never wetted
Insulation	Silica aerogel 3.0; silicone foam 2.0	core, sleeve	thin, 200 °C+

NOTES

1. All wiring silicone-insulated 0.75 mm² (18 AWG), 200 °C rated, inside the core; PVC allowed only in the base.
2. Heater loop current 3 A at 20 V; fuse 5 A fast-blow; PD trigger set to 20 V (solder jumper or switch).
3. No PCB: the wax gate (P14) and the overflow standpipe (P11) provide all process control.
4. Earth/drain: shell P03 bonded to supply –ve via 1 MΩ + 10 nF (ESD); 20 V SELV, no mains inside.

MPx-BLM-020 · Wiring, seals, fasteners, material spec

MATERIAL	FINISH	SCALE
see tables	—	NTS
QTY / UNIT	MASS (est.)	UNITS · TOL.
1	60 g (electrics)	mm · ISO 2768-m
PROJECTION	DRAWN	DATE
First angle (IS/ISO)	MPx team (PD-298)	21 Oct 2026